Differential Geometry

Exercise 11

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11.1 Problem (DoCarmo, Riemannian Geometry, page 46 question 7)

Let G be a compact connected Lie group of dimension n.

(1) Show that if ω is a left-invariant n-form, then it is right-invariant.

(2) Show that there exists a non-vanishing left-invariant n-form ω on G.

(3) Let $\langle \bullet, \bullet \rangle$ be a left invariant metric on G, and ω a positive n-form on G which is left-invariant. Define a new Riemannian metric $\langle\langle \bullet, \bullet \rangle\rangle$ on G by

$$\langle\langle u, v \rangle\rangle_y = \int_G \langle (dR_x)_y u, (dR_x)_y v \rangle_{yx} \omega_x, \qquad x, y \in G,\ u, v \in T_y G.$$

Show that this is a bi-invariant metric.

(1) Let $a \in G$, then $R_a^* \omega$ is left-invariant as for $x \in G$ we have

$$L_x^* R_a^* \omega = R_a^* L_x^* \omega = R_a^* \omega.$$

All left-invariant forms are determined by their values at the identity, and since $\Lambda^n T_e^* G$ is one-dimensional, this means that all left-invariant n-forms are scalar multiples of one another. Thus $R_a^* \omega = f(a)\omega$ for some $f(a) \in \mathbb{R} - 0$ (since ω is non-vanishing, so too is $R_a^* \omega$ since R_a is a diffeomorphism).

Now,

$$R_{ba}^* \omega = R_a^*(R_b^* \omega) = R_a^*(f(b)\omega) = f(b) R_a^* \omega = f(b) f(a) \omega,$$

which means that $f(ba) = f(b)f(a)$ so f is a group morphism from G to the multiplicative group of reals $\mathbb{R} - 0$. Furthermore it is continuous, as we can compute $f(a)$ by taking a basis $v_1, \dots, v_n \in T_e G$ then

$$f(a) = \frac{R_a^* \omega(v_1, \dots, v_n)}{\omega(v_1, \dots, v_n)},$$

and $a \mapsto R_a^* \omega$ is smooth, so this is just scaling a smooth map.

Since G is compact and connected, $f(G) \subseteq \mathbb{R} - 0$ is then a compact connected subgroup. It is thus a closed interval $[c, d]$, which means $c^2 \geq c, d^2 \leq d$ which implies $d \leq 1 \leq c$ so $f(G) = \{1\}$. Thus $R_a^* \omega = \omega$ for all $a \in G$, and ω is right-invariant.

(2) Let (E_i) be a left-invariant global frame of TG, i.e. left-invariant vector fields obtained from a basis of $T_e G$. Take (ε^j) its dual global coframe, then define

$$\omega = \varepsilon^1 \wedge \cdots \wedge \varepsilon^n.$$

Then for $a \in G$ we have

$$L_a^* \omega = (L_a^* \varepsilon^1) \wedge \cdots \wedge (L_a^* \varepsilon^n).$$

Now we have

$$L_a^* \varepsilon^j (E_i) = \varepsilon^j (L_a^* E_i) = \varepsilon^j (E_i) = \delta_i^j,$$

which shows that $L_a^* \varepsilon^j = \varepsilon^j$, meaning $L_a^* \omega = \omega$, so ω is left-invariant and non-vanishing.

(3) Since all non-vanishing n-forms on a connected manifold are either positive or negative. This is because if we take a positive n-form ω, all n-forms are multiples of this by a smooth function $u\omega$.

If they are nonvanishing, then u is never zero. If the manifold is connected, this means u is either strictly positive or strictly negative.

So let ω be a nonvanishing left-invariant n-form. By potentially multiplying by -1, we can assume it is positive. Then we define $\langle\!\langle \bullet, \bullet \rangle\!\rangle$ as in the question.

By linearity of the differential, it is immediate that $\langle\!\langle \bullet, \bullet \rangle\!\rangle$ is bilinear. It is also symmetric by symmetry of $\langle \bullet, \bullet \rangle$. Furthermore, $\langle (dR_x)_y u, (dR_x)_y u \rangle_{yx} \omega$ is a positive n-form when $u \neq 0$ (since the inner product is positive), and thus its integral is positive. Thus for $u \neq 0$, $\langle\!\langle u, u \rangle\!\rangle > 0$. For $u = 0$, the integral is zero, so $\langle\!\langle u, u \rangle\!\rangle = 0$.

Now, we have

$$\begin{aligned}
\langle\!\langle d(L_a)_y u, d(L_a)_y v \rangle\!\rangle_{ay} &= \int_G \langle (dR_x)_{ay} d(L_a)_y u, (dR_x)_{ay} d(L_a)_y v \rangle_{ayx} \omega \\
&= \int_G \langle d(R_x \circ L_a)_y u, d(R_x \circ L_a)_y v \rangle_{ayx} \omega \\
&= \int_G \langle d(L_a \circ R_x)_y u, d(L_a \circ R_x)_y v \rangle_{ayx} \omega \\
&= \int_G \langle d(L_a)_{yx} d(R_x)_y u, d(L_a)_{yx} d(R_x)_y v \rangle_{ayx} \omega \\
&= \int_G \langle d(R_x) u, d(R_x)_y v \rangle_{yx} \omega \\
&= \langle\!\langle u, v \rangle\!\rangle_y \,,
\end{aligned}$$

where the final equality is due to the left-invariance of $\langle \bullet, \bullet \rangle$. So we see that $\langle\!\langle \bullet, \bullet \rangle\!\rangle$ remains left-invariant.

Now we claim right-invariance. Let us write $f_{u,v} \colon G \to \mathbb{R}$ for the map $f_{u,v}(x) = \langle (dR_x)_y u, (dR_x)_y v \rangle$. Thus

$$\langle\!\langle u, v \rangle\!\rangle_y = \int_G f_{u,v} \omega.$$

And we have

$$L_a^*(f_{u,v}\omega) = (f_{u,v} \circ L_a)(L_a^* \omega) = (f_{u,v} \circ L_a)\omega$$

since ω is left-invariant. Furthermore, for all $a \in G$, L_a is orientation-preserving. This is because $L_a^* \omega = \omega$, and ω is the orientation form.

Thus we have

$$\begin{aligned}
\langle\!\langle d(R_a)_y u, d(R_a)_y v \rangle\!\rangle_{ya} &= \int_G \langle (dR_x)_{ya} d(R_a)_y u, (dR_x)_{ya} d(R_a)_y v \rangle_{yax} \omega \\
&= \int_G \langle d(R_x \circ R_a)_y u, d(R_x \circ R_a)_y v \rangle_{yax} \omega \\
&= \int_G \langle d(R_{ax})_y u, d(R_{ax})_y v \rangle_{yax} \omega \\
&= \int_G (f_{u,v} \circ L_a) \omega \\
&= \int_G L_a^*(f_{u,v}\omega) \\
&= \int_G f_{u,v} \omega \\
&= \langle\!\langle u, v \rangle\!\rangle_y \,.
\end{aligned}$$

Showing right-invariance, as desired. G being compact is necessary for these integrals to be defined, as otherwise ω may not be compactly supported.

Finally smoothness follows from left-invariance. This is because for any vector fields X, Y, by left invariance

$$\langle\!\langle X, Y\rangle\!\rangle_a = \langle\!\langle X_a, Y_a\rangle\!\rangle_a = \langle\!\langle d(L_{a^{-1}})X_a, d(L_{a^{-1}})Y_a\rangle\!\rangle_e\,,$$

and $a \mapsto d(L_{a^{-1}})X_a$ is smooth as a composition of smooth functions. Then inner product is taken in the identity which is smooth. Thus we have constructed a bi-invariant Riemannian metric.

11.2 Problem

Let (M, g) be Riemannian. For a smooth real-valued function $f\colon M \to \mathbb{R}$ we define its **gradient** to be the unique vector field ∇f which satisfies $g(\nabla f, X) = df(X) = Xf$ for all vector fields X. The **Hessian** of f is the covariant 2-tensor defined by

$$\mathrm{Hess} f(X, Y) = g(\nabla_X \nabla f, Y).$$

For $p \in M$ define $f_p\colon M \to \mathbb{R}$ by $f_p(q) = \frac{1}{2} d(p, q)^2$.

(1) Show that for $f\colon M \to \mathbb{R}$, $\mathrm{Hess} f$ is symmetric.

(2) Suppose there is a constant $c \geq 0$ such that $\mathrm{Hess} f(v, v) \geq c g(v, v)$ for all tangent vectors v. Let γ be a unit speed geodesic. Show that

$$\frac{d^2}{dt^2}(f \circ \gamma) \geq c.$$

In particular $f \circ \gamma$ is convex, and strictly convex for $c > 0$.

(3) Let B be a normal ball around p. Show that f_p is smooth on B. Let $\gamma\colon [0, 1] \to B$ be a geodesic such that $\gamma(0) = p$ and $\gamma(1) = q$. Show that $\nabla f_p(q) = \dot\gamma(q)$.

(4) Continuing with the case in the previous point, let $v, w \in T_pM$ be such that $\exp_p(v) = \gamma(1)$. For sufficiently small $\varepsilon > 0$ and $(s, t) \in (-\varepsilon, \varepsilon) \times [0, 1]$, let

$$\overline{\gamma}(s, t) = \overline{\gamma}_s(t) = \exp_p(t(v + sw)).$$

Then define

$$J = \left.\frac{d}{ds}\right|_{s=0} \overline{\gamma}$$

to be the Jacobi field along $\gamma = \overline{\gamma}_0$ such that $J(0) = 0$ and $J'(0) = w$. Show that

$$\mathrm{Hess} f_p(J(1), J(1)) = g(J'(1), J(1)).$$

(5) Again we continue with the previous point. Assume the sectional curvature satisfies $K \leq 0$. For $t > 0$, let

$$\lambda(t) = \frac{g(J'(t), J(t))}{|J(t)|^2}.$$

Prove that $\lambda(t)$ is well-defined and strictly positive for $t > 0$. Prove that $\dot\lambda(t) \geq -\lambda^2$, and show this is strict when $K < 0$ and v, w are linearly independent.

(6) Again we continue with the previous point. Show that $\lambda(t) \to \infty$ as $t \to 0$. Using the previous point, show that $\frac{\dot\lambda}{\lambda^2} + 1$ is well-defined and non-negative for $t > 0$. Integrate over $t \in [0, 1]$ to obtain $\lambda(1) \geq 1$.

(7) Let M have nonpositive sectional curvature and $B \subseteq M$ be a normal ball centered at p. Conclude from the previous point that for all $q \in B$ and $v \in T_qM$, $\mathrm{Hess} f_p(v,v) \geq g(v,v)$. Show that if $K < 0$ and $v, \nabla f_p(q)$ are linearly independent, then this inequality is strict.

(8) Let M be complete and simply connected, with $K \leq 0$. Show that f_p is smooth on all of M.

(9) Continuing the previous point, let $\varphi\colon M \to M$ be an isometry of finite order k. Show that φ has a fixed point as follows: for $p \in M$ let

$$h = \sum_{i=0}^{k-1} f_{\varphi^i(p)}.$$

Show that $h \circ \varphi = h$, and then from point (7) conclude $\mathrm{Hess} h > 0$. Using Hopf-Rinow and part (2), show that h has a unique minimum, which must be a fixed point of φ.

(10) Continue with point (8). Let $p \in M$ and $\gamma\colon [0,t] \to M$ be a geodesic. Combining points (2) and (7), show that

$$\frac{d^2}{dt^2}(f_p \circ \gamma) \geq 1,$$

and integrating twice we obtain

$$d(p,\gamma(t))^2 \geq d(p,\gamma(0))^2 + 2g(\nabla f_p(\gamma(0)), \dot\gamma(0))t + t^2.$$

Using part (3), interpret this as a generalized law of cosines. In particular for any geodesic triangle in M with side lengths a, b, c, and angle α opposite side a, we have

$$a^2 \geq b^2 + c^2 - 2bc\cos\alpha.$$

Show that this inequality is strict for $K < 0$ and a non-degenerate triangle.

(11) Continue with part (10). Show that the sum of angles in a geodesic triangle in M is less than or equal to π. Furthermore this inequality is strict for $K < 0$ and a nondegenerate triangle.

(1) We have by metric-compatibility

$$\mathrm{Hess} f(X,Y) = g(\nabla_X \nabla f, Y) = X(g(\nabla f, Y)) - g(\nabla f, \nabla_X Y) = X(Yf) - \nabla_X Y(f).$$

And thus

$$\mathrm{Hess} f(Y,X) = Y(Xf) - \nabla_Y X(f).$$

Subtracting both sides gives

$$\mathrm{Hess} f(X,Y) - \mathrm{Hess} f(Y,X) = X(Yf) - Y(Xf) - (\nabla_X Y(f) - \nabla_Y X(f)),$$

and both terms are $[X,Y]f$: the first by definition and the second by symmetry of the connection. Thus $\mathrm{Hess} f(X,Y) = \mathrm{Hess} f(Y,X)$ as desired.

(2) For t_0 in γ's domain, let $p = \gamma(t_0)$ and consider a geodesic frame centered at p, (E_i). Then (E_i) are orthonormal, and so the components of ∇f with respect to this local frame are given by

$$g(\nabla f, E_i) = df(E_i) = E_i(f).$$

Thus

$$\nabla f = \sum_i (E_i f) E_i.$$

Let us also represent $\dot\gamma$ in terms of this geodesic frame: $\dot\gamma = a^i(t)E_i(\gamma(t))$. Then we have

$$\begin{aligned}\mathrm{Hess}f(\dot\gamma(t),\dot\gamma(t)) &= g\left(\nabla_{a^iE_i(\gamma)}\sum_j (E_jf)E_j, a^kE_k(\gamma)\right)\\ &= a^ia^kg\left(\nabla_{E_i}\sum_j(E_jf)E_j, E_k\right)_\gamma\\ &= \sum_j a^ia^kg(E_i(E_jf)E_j + (E_jf)\nabla_{E_i}E_j, E_k).\end{aligned}$$

Evaluating at t_0, this becomes

$$= \sum_j a^ia^kg(E_i(E_jf)E_j, E_k) = a^i(t_0)a^j(t_0)E_i(E_jf)|_p.$$

On the other hand,

$$\frac{d}{dt}(f\circ\gamma) = df(a^i(t)E_i(\gamma(t))) = a^i(t)(E_if)\circ\gamma.$$

Differentiating again gives

$$\frac{d^2}{dt^2}(f\circ\gamma) = \dot a^i(t)(E_if)\circ\gamma + a^i(t)d(E_if)(a^jE_j|_\gamma) = \dot a^i(E_if)|_\gamma + a^ia^jE_j(E_if)|_\gamma.$$

Notice that the first term is just $D_t\dot\gamma = 0$, so we see that

$$\frac{d^2}{dt^2}(f\circ\gamma) = a^ia^jE_i(E_jf)|_\gamma,$$

which is equal to $\mathrm{Hess}f(\dot\gamma,\dot\gamma)$ at t_0.

This is true for all t_0, so we see that

$$\frac{d^2}{dt^2}(f\circ\gamma) = \mathrm{Hess}f(\dot\gamma,\dot\gamma) \geq cg(\dot\gamma,\dot\gamma) = c\,|\dot\gamma|^2 = c$$

as desired.

(3) Let $\widetilde{B} \subseteq T_pM$ be the tangent ball mapped diffeomorphically to B by $\exp_p$. Radial geodesics are minimizing, so for $v \in \widetilde{B}$ we have $d(p, \exp_p(v)) = |v|$. Thus

$$f_p(\exp_p(v)) = \frac{1}{2}|v|^2.$$

Thus $f_p \circ \exp_p$ is smooth, and since $\exp_p$ is a diffeomorphism $\widetilde{B} \to B$, we see that f_p is smooth on B.

Let $\overline{g}$ denote the Euclidean metric on T_pM, so $\exp_p\colon \widetilde{B} \to B$ is an isometry. In general if we have an isometry $\varphi\colon (N,\widetilde{g}) \to (M,g)$ and $f\colon M \to \mathbb{R}$, then

$$\widetilde{g}(\nabla(f\circ\varphi), X) = d(f\circ\varphi)(X) = df(\varphi_*X) = g(\nabla f, \varphi_*X).$$

Since φ_* is an isometry, this is equal to

$$= \widetilde{g}(\varphi_*^{-1}\nabla f, X).$$

This is true for all X, so

$$\nabla(f\circ\varphi) = \varphi_*^{-1}\nabla f \Longrightarrow \nabla f = \varphi_*\nabla(f\circ\varphi).$$

In our case,

$$\nabla f_p = (\exp_p)_* \nabla (f \circ \exp_p),$$

and $f \circ \exp_p(v) = |v|^2/2$ whose gradient is then just v (gradients in the Euclidean case are computed classically). The geodesic connecting p and q has the form $\gamma(t) = \exp_p(tv)$ for $q = \exp_p v$, then

$$\nabla f_p(q) = \nabla f_p(\exp_p v) = d(\exp_p)v = \dot{\gamma}(1).$$

(4) Notice that

$$J'(1) = \partial_t \partial_s \overline{\gamma}(0,1) = \partial_s \partial_t \overline{\gamma}(0,1) = \partial_s \dot{\overline{\gamma}}_s(1)\big|_{s=0}$$

where the final equality is due to symmetry. By the previous point

$$\nabla f_p(\overline{\gamma}_s(1)) = \dot{\overline{\gamma}}_s(1).$$

And by definition we have

$$\mathrm{Hess} f_p(J(1), J(1)) = g(\nabla_{J(1)} \nabla f_p, J(1)).$$

Consider the first term:

$$\nabla_{J(1)} \nabla f_p = \nabla_{\partial_s \overline{\gamma}(0,1)} \nabla f_p,$$

which is equal to the covariant derivative of $\nabla f_p(\overline{\gamma}_s(1))$ with respect to s taken along $\overline{\gamma}_s(1)$, evaluated at 0. That is, it is equal to

$$D_s \nabla f_p(\overline{\gamma}_s(1))\big|_{s=0} = D_s \dot{\overline{\gamma}}_s(1)\big|_{s=0} = J'(1),$$

where the final equality is due to the first equation we showed. Thus we have

$$\mathrm{Hess} f_p(J(1), J(1)) = g(J'(1), J(1))$$

as desired.

(5) Since M has nonpositive sectional curvature, no two points are conjugate along any geodesic. Since $J(0) = 0$, this means that $J(t) \neq 0$ for all $t \neq 0$. Thus λ is well-defined.

Computing the derivative of λ, we get

$$\dot{\lambda} = \frac{(g(J'', J) + g(J', J'))\, |J|^2 - 2g(J', J)^2}{g(J, J)^2},$$

and

$$\lambda^2 = \frac{g(J', J)^2}{g(J, J)^2}.$$

So $\dot{\lambda} \geq -\lambda^2$ iff

$$(g(J'', J) + |J'|^2)\, |J|^2 \geq g(J', J)^2.$$

Cauchy-Schwarz gives us

$$g(J', J) \leq |J'|^2 |J|^2$$

and so if we can show that $g(J'', J)$ is nonnegative we will have the inequality.

By the Jacobi equation

$$g(J'', J) = g(-R(\dot{\gamma}, J)\dot{\gamma}, J) = -\mathrm{Rm}(\dot{\gamma}, J, \dot{\gamma}, J).$$

If $\dot{\gamma}, J$ are linearly dependent this is zero and we are finished. Otherwise this is just a negative multiple of the sectional curvature at $\dot{\gamma}, J$. Since the sectional curvature is nonpositive, this is nonnegative and we are finished.

Notice that this also implies $\lambda(t) > 0$. This is because as we showed,

$$g(J', J)' = |J'|^2 + g(J'', J) \geq 0.$$

Thus $g(J'(t), J(t))$ is nondecreasing, and at t it is zero. Thus it is nonnegative for $t > 0$, and since at $t = 0$ we have $|J'|^2 = |w|^2 > 0$ it is strictly increasing in a neighborhood of 0, so $g(J', J) > 0$ for $t > 0$. Thus $\lambda(t) > 0$ for $t > 0$ as desired.

Now if $K < 0$ and v, w are linearly independent, we want to show this inequality is strict. In lieu of our previous analysis, it is sufficient to show that $\dot{\gamma}(t), J(t)$ is linearly independent. We know that, since $J(0) = 0$, $J'(0) = w$, and $\dot{\gamma}(0) = v$,

$$g(J(t), \dot{\gamma}(t)) = g(J'(0), \dot{\gamma}(0))t + g(J(0), \dot{\gamma}(0)) = g(w, v)t.$$

If J and $\dot{\gamma}$ were dependent, this would mean

$$J = \frac{g(J, \dot{\gamma})}{g(\dot{\gamma}, \dot{\gamma})}\dot{\gamma} = \frac{g(w, v)t}{g(v, v)}\dot{\gamma}.$$

Let us denote the right-hand side by $\widehat{J}$. This is a Jacobi field because fields of the form $at\dot{\gamma}(t)$ are Jacobi fields: their first derivative is $a\dot{\gamma}$, and so their second is zero. Then $R(at\dot{\gamma}, \dot{\gamma})\dot{\gamma} = 0$, so they are Jacobi fields.

Since Jacobi fields form a vector space, this means $J - \widehat{J}$ is a Jacobi field. But $J(0) = \widehat{J}(0) = 0$ and they also agree at another point t. But since M has nonpositive sectional curvature, no two points are conjugate along any geodesic. This means that $J - \widehat{J}$ cannot be a nontrivial Jacobi field, as otherwise $\gamma(0), \gamma(t)$ would be conjugate. Thus $J = \widehat{J}$, but since v, w are linearly independent

$$\widehat{J}'(0) = \frac{g(w, v)}{g(v, v)}v \neq w = J'(0).$$

(6) We note that as $t \to 0$, both the numerator and denominator approach 0 since $J(t)$ is smooth and $J(0) = 0$. Thus by Lhopital, the limit is equal to

$$\lim_{t \to 0} \lambda(t) = \lim_{t \to 0} \frac{g(J'', J) + g(J', J')}{2g(J', J)}.$$

The numerator of the limit approaches $g(J'(0), J'(0)) > 0$, while the denominator approaches 0. Thus the limit is infinite (everything in the limit is positive, so this converges to $+\infty$), as desired.

Since $\lambda(t) > 0$ for $t > 0$, $\dot{\lambda}/\lambda^2 + 1$ is well-defined for $t > 0$. Since $\dot{\lambda} \geq -\lambda^2$, we see that $\dot{\lambda}/\lambda^2 \geq -1$ so this is nonnegative for $t > 0$. Integrating yields

$$\int_0^1 \frac{\dot{\lambda}}{\lambda^2} + 1\, dt = -\frac{1}{\lambda}\bigg|_0^1 + 1 = -\frac{1}{\lambda(1)} + 1,$$

as we showed the limit of $\lambda(t)$ as $t \to 0$ is infinite, and thus $1/\lambda \to 0$. Since the integrand is nonnegative, we see

$$-\frac{1}{\lambda(1)} + 1 \geq 0 \Longrightarrow \frac{1}{\lambda(1)} \leq 1 \Longrightarrow \lambda(1) \geq 1$$

as desired.

Note that if $K < 0$ and v, w are linearly independent then the inequality $\dot{\lambda} > -\lambda^2$ is strict. Thus the integrand is strictly positive and so we get that

$$-\frac{1}{\lambda(1)} + 1 > 0 \Longrightarrow \lambda(1) > 1.$$

(7) From (4) we have that

$$\mathrm{Hess}f_p(J(1),J(1)) = g(J'(1),J(1)) = \lambda(1)g(J(1),J(1)) \geq g(J(1),J(1)).$$

For $q \in B$ and $v \in T_qM$, we want a Jacobi field J such that $J(1) = v$.

Let us write $q = \exp_p(u)$, so we want $x, y \in T_pM$ such that $J = \partial_s \exp_p(t(x+sy))|_{s=0}$ maps $J(1) = v$. Since

$$J(1) = \left.\frac{\partial}{\partial s}\right|_{s=0} \exp_p(x+sy) = d(\exp_p)_x(y),$$

we need $\exp_p(x) = q$, meaning we need $x = u$. Then since $\exp_p$ is a diffeomorphism to B, there will exist a y which satisfies this. So this gives us a Jacobi field J where $J(1) = v$, and thus

$$\mathrm{Hess}f_p(v,v) \geq g(v,v).$$

This inequality is strict when x, y are linearly independent, as then $\lambda(1) > 1$. Since $x = u$ and $y = d(\exp_p^{-1})_u(v)$, this is iff $v = d\exp_p(y)$ and $d\exp_p(u)$ are linearly independent. Now,

$$\nabla f_p(q) = \dot\gamma(1),$$

where γ is the geodesic connecting p and q, so $\gamma(t) = \exp_p(tu)$. Then

$$\nabla f_p(q) = d\exp_p(u),$$

so x, y are linearly independent iff $v, \nabla f_p(q)$ are linearly independent as desired.

(8) By Hadamard, $\exp_p\colon T_pM \to M$ is a diffeomorphism. This means that M is itself a normal ball, and as we showed f_p is smooth on normal balls. Thus f_p is smooth, as desired.

(9) Notice that since φ is an isometry,

$$f_{\varphi^i(p)} \circ \varphi(q) = \frac{1}{2}d(\varphi^i(p), \varphi(q))^2 = \frac{1}{2}d(\varphi^{i-1}(p), q) = f_{\varphi^{i-1}(p)}(q).$$

Thus $f_{\varphi^i(p)} \circ \varphi = f_{\varphi^{i-1}(p)}$ and so

$$h \circ \varphi = \sum_{i=0}^{k-1} f_{\varphi^i(p)} \circ \varphi = \sum_{i=0}^{k-1} f_{\varphi^{i-1}(p)} = h.$$

Now notice that given two smooth real-valued functions f, f', we have $\nabla(f+f') = \nabla f + \nabla f'$. This is because $d(f+f') = df + df'$, and the musical isomorphisms being linear. Thus

$$\mathrm{Hess}(f+f')(X,Y) = g(\nabla_X \nabla(f+f'), Y) = g(\nabla_X(\nabla f + \nabla f'), Y) = \\ g(\nabla_X \nabla f, Y) + g(\nabla_X \nabla f', Y) = \mathrm{Hess}f(X,Y) + \mathrm{Hess}f'(X,Y).$$

Thus $\mathrm{Hess}(f+f') = \mathrm{Hess}f + \mathrm{Hess}f'$.

And thus

$$\mathrm{Hess}h = \sum_{i=0}^{k-1} \mathrm{Hess}(f_{\varphi^i(p)}).$$

Since $\mathrm{Hess}f_p(v,v) \geq g(v,v) > 0$ for all p, we see that $\mathrm{Hess}h \geq kg(v,v) > 0$ as desired.

Now suppose h had two minima, $p \neq q$. Since M is complete, there is a minimizing unit geodesic γ from p to q, so $p = \gamma(0)$ and $q = \gamma(a)$ say. Then from (2) we have

$$\frac{d^2}{dt^2}(h \circ \gamma) \geq k > 0,$$

which means that $h \circ \gamma$ is strictly convex, and thus can only have a single minimum, contradicting both $h \circ \gamma(0)$ and $h \circ \gamma(a)$ both being minima.

Now since $h(p) \geq 0$, h's image is bounded from below. Let $c = \inf h(M)$, and let $x_n \in M$ be a sequence such that $h(x_n) \to c$. For sufficiently large n, we then have $h(x_n) < c + 1$, and since $d(p,x)^2 \leq h(x)$ we have $d(p, x_n) \leq \sqrt{c+1}$. Thus x_n eventually lies within the closed metric ball around p of radius $\sqrt{c+1}$. This is bound and closed, so by Hopf-Rinow, compact. Thus x_n has a convergent subsequence x_{m_n}, say $x_{m_n} \to x_0$.

Since $h(x_n) \to c$, we have that $h(x_{m_n}) \to c$ and by continuity $h(x_{m_n}) \to h(x_0)$. Thus $h(x_0) = c$, and since c is the infimum of the image of h, x_0 is a minimum of h. As we showed the minimum is unique, it is the only minimum.

Now, $h \circ \varphi = h$ means that $h(\varphi(x_0)) = h(x_0)$ is minimal. Since h has a unique minimum, this means $\varphi(x_0) = x_0$ so x_0 is a fixed point of φ.

(10) From (7) we have $\text{Hess} f_p(v,v) \geq g(v,v)$ which means by (2) that

$$\frac{d^2}{dt^2}(f_p \circ \gamma) \geq 1$$

for all unit velocity geodesics γ. Integrating once we see that

$$\frac{d}{dt} f_p \circ \gamma(t) = \int_0^t \frac{d^2}{dt^2}(f_p \circ \gamma) + \frac{d}{dt} f_p \circ \gamma(0) \geq t + \frac{d}{dt} f_p \circ \gamma(0).$$

Integrating again gives

$$f_p \circ \gamma(t) \geq \frac{t^2}{2} + t \frac{d}{dt} f_p \circ \gamma(0) + f_p \circ \gamma(0).$$

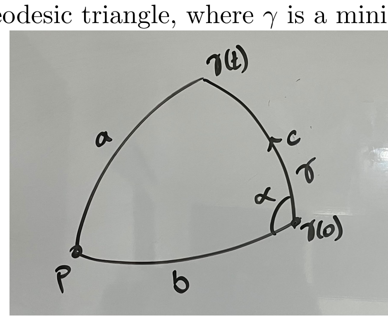
Recall that

$$g(\nabla f_p(\gamma(0)), \dot{\gamma}(0)) = df_p(\dot{\gamma}(0)) = \frac{d}{dt} f_p \circ \gamma(0).$$

Thus multiplying the above inequality by two, we get the desired

$$d(p, \gamma(t))^2 \geq d(p, \gamma(0))^2 + 2g(\nabla f_p(\gamma(0)), \dot{\gamma}(0))t + t^2.$$

Consider the following geodesic triangle, where γ is a minimizing unit-velocity geodesic



Then $a^2 = d(p, \gamma(t))^2$, $b = d(p, \gamma(0))^2$, and $c = L(\gamma)^2 = t^2$. If we let β be the geodesic from $\beta(0) = \gamma(0)$ to $\beta(1) = p$, then we know that $\nabla f_p(\gamma(0)) = -\dot{\beta}(0)$ (since $\beta(1-t)$ is the geodesic in the reverse direction). Thus

$$g(\nabla f_p(\gamma(0)), \dot{\gamma}(0)) = -g(\dot{\beta}(0), \dot{\gamma}(0)).$$

The angle α is given by the angle between $\dot{\beta}(0)$ and $\dot{\gamma}(0)$, so

$$\cos \alpha = \frac{g(\dot{\beta}(0), \dot{\gamma}(0)}{\left|\dot{\beta}\right|}.$$

And since β is a geodesic, $b = L(\beta) = \left|\dot{\beta}\right|$. Thus

$$2g(\nabla f_p(\gamma(0)), \dot{\gamma}(0))t = -2bt\cos\alpha.$$

Since $c = t$ we see that this is equal to $-2bc\cos\alpha$.

Thus we have

$$a^2 \geq b^2 + c^2 - 2bc\cos\alpha$$

as desired.

This inequality is strict when $K < 0$ and the triangle is non-degenerate because then the inequality $\frac{d^2}{dt^2}(f_p \circ \gamma) > 1$ is strict. Then integrating preserves this strict inequality.

(11) Let T be the triangle in M, with side lengths a, b, c and associated angles α, β, γ. By the triangle inequality, $c \leq a + b$ so there exists a Euclidean triangle T' with the same side lengths. Let the associated angles be α', β', γ'. Then we have

$$a^2 = b^2 + c^2 - 2bc\cos\alpha' \geq b^2 + c^2 - 2bc\cos\alpha.$$

This implies that $\cos\alpha \geq \cos\alpha'$. Since in $[0, \pi]$ cosine is decreasing, this means that $\alpha' \geq \alpha$. The same holds for β, γ.

Thus we have

$$\alpha + \beta + \gamma \leq \alpha' + \beta' + \gamma' = \pi$$

as desired. When $K < 0$ and the triangle is non-degenerate, then the inequality above is strict: $\cos\alpha > \cos\alpha'$. Thus $\alpha < \alpha'$ and the inequality of the sum is strict.